

第十三章 积分变换法

本章基本要求

1. 掌握用傅里叶变换法求解无界及半无界空间的定解问题的方法;
2. 掌握用拉普拉斯变换法求解含时间的定解问题的方法。

§ 13.1 傅里叶变换法

1. 求解无限长传输线上的电振荡传播问题。 $G:C=R:L$ 的情形跟 $G:C \neq R:L$ 的情形有什么不同?

解: 电报方程 (7.1.12) 及 (7.1.13) 中, 电流电压满足同样的方程, 现以 u 统一代表电流和电压, 则 u 满足方程

$$LCu_{tt} - u_{xx} + (LG + RC)u_t + RG u = 0.$$

对方程中 x 变数作傅里叶变换, $u(x, t) \Rightarrow U(k, t)$, 得

$$LCU'' + (LG + RC)U' + (k^2 + RG)U = 0.$$

即

$$LCU'' + \frac{LG + RC}{LC}U' + \frac{k^2 + RG}{LC}U = 0.$$

该方程的解是

$$U(t; k) = A(k) \exp \left[-\frac{R}{2L}t - \frac{G}{2C}t + \frac{1}{2} \sqrt{\left(\frac{R}{L} - \frac{G}{C} \right)^2 - \frac{4k^2}{LC}} t \right] \\ + B(k) \exp \left[-\frac{R}{2L}t - \frac{G}{2C}t - \frac{1}{2} \sqrt{\left(\frac{R}{L} - \frac{G}{C} \right)^2 - \frac{4k^2}{LC}} t \right].$$

做逆变换

$$u(x, t) = e^{-\frac{R}{2L}t - \frac{G}{2C}t} \int_{-\infty}^{\infty} \left[A(k) e^{\frac{1}{2} \sqrt{\left(\frac{R}{L} - \frac{G}{C} \right)^2 - \frac{4k^2}{LC}} t} + B(k) e^{-\frac{1}{2} \sqrt{\left(\frac{R}{L} - \frac{G}{C} \right)^2 - \frac{4k^2}{LC}} t} \right] e^{ikx} dk \quad (1)$$

讨论:

(i) 如 $\frac{R}{L} = \frac{G}{C}$, 这时①式成为

$$u(x, t) = e^{-\frac{R}{2L}t - \frac{G}{2C}t} \int_{-\infty}^{\infty} \left[A(k) e^{i \frac{k}{\sqrt{LC}} t} + B(k) e^{-i \frac{k}{\sqrt{LC}} t} \right] e^{ikx} dk,$$

这是沿正负方向传播的衰减波动, 通过的频率无限制, 并且无色散。

(ii) $\frac{R}{L} \neq \frac{G}{C}$, 若 $\left(\frac{R}{L} - \frac{G}{C} \right)^2 - \frac{4k^2}{LC} < 0$, 即 $|k| > \frac{\sqrt{LC}}{2} \left| \frac{R}{L} - \frac{G}{C} \right|$ 时, 有衰减波动。波速 $= \frac{1}{2} \sqrt{\frac{4}{LC} - \frac{1}{k^2} \left(\frac{R}{L} - \frac{G}{C} \right)^2}$, 其随 k 的不同而不同, 有色散。

(iii) $\frac{R}{L} \neq \frac{G}{C}$, 若 $\left(\frac{R}{L} - \frac{G}{C} \right)^2 - \frac{4k^2}{LC} > 0$, 即 $|k| < \frac{\sqrt{LC}}{2} \left| \frac{R}{L} - \frac{G}{C} \right|$ 时, (5) 式中不出现虚数项,

这时只有单纯衰减而不振荡。

因此, 传输线相当于高通滤波器, 只通过 $|k| > \frac{\sqrt{LC}}{2} \left| \frac{R}{L} - \frac{G}{C} \right|$ 的振荡。

2. 研究半无限长细杆导热问题. 杆端 $x=0$ 温度保持为零. 初始温度分布为 $K(e^{-\lambda x} - 1)$.

解: 定解问题是

$$\begin{cases} u_t - a^2 u_{xx} = 0, \\ u|_{x=0} = 0, \\ u|_{t=0} = K(e^{-\lambda x} - 1). \end{cases} \quad (x \neq 0) \quad (1)$$

首先将问题奇延拓 (第一类齐次边值) 到 $x < 0$ 的半空间。

$$\begin{cases} u_t - a^2 u_{xx} = 0, \\ u|_{t=0} = \varphi(x) = \begin{cases} K(e^{-\lambda x} - 1) & (x > 0) \\ -K(e^{\lambda x} - 1) & (x < 0) \end{cases} \end{cases} \quad (2)$$

对②作傅里叶变换, 定解问题变换为

$$\begin{cases} U' + k^2 a^2 U = 0, \\ U|_{t=0} = \Phi(k). \end{cases}$$

这个常微分方程的初始值问题的解是

$$U(t, k) = \Phi(k) e^{-k^2 a^2 t}.$$

再进行逆傅里叶变换,

$$\begin{aligned} u(x, t) &= \mathcal{F}^{-1}[U(t, k)] = \int_{-\infty}^{\infty} \Phi(k) e^{-k^2 a^2 t} e^{ikx} dk \\ &= \frac{1}{2\pi} \int_{-\infty}^{\infty} \left[\int_{-\infty}^{\infty} \varphi(\xi) e^{-ik\xi} d\xi \right] e^{-k^2 a^2 t} e^{ikx} dk \end{aligned}$$

交换积分次序

$$u(x, t) = \frac{1}{2\pi} \int_{-\infty}^{\infty} \varphi(\xi) \left[\int_{-\infty}^{\infty} e^{-k^2 a^2 t} e^{ik(x-\xi)} dk \right] d\xi.$$

引用教材本节例 2 的积分公式 $\int_{-\infty}^{\infty} e^{-\alpha^2 k^2} e^{\beta k} dk = (\sqrt{\pi}/\alpha) e^{\beta^2/4\alpha^2}$ 计算 [] 内的积分,

置 $\alpha = a\sqrt{t}, \beta = i(x-\xi)$ 以利用此积分公式, 即得

$$\begin{aligned} u(x, t) &= \int_{-\infty}^{\infty} \varphi(\xi) \left[\frac{1}{2a\sqrt{\pi t}} e^{-\frac{(x-\xi)^2}{4a^2 t}} \right] d\xi \\ &= \int_{-\infty}^0 [-K(e^{\lambda \xi} - 1)] \left[\frac{1}{2a\sqrt{\pi t}} e^{-\frac{(x-\xi)^2}{4a^2 t}} \right] d\xi + \int_0^{\infty} [K(e^{-\lambda \xi} - 1)] \left[\frac{1}{2a\sqrt{\pi t}} e^{-\frac{(x-\xi)^2}{4a^2 t}} \right] d\xi \\ &= \int_{-\infty}^0 K \left[\frac{1}{2a\sqrt{\pi t}} e^{-\frac{(x-\xi)^2}{4a^2 t}} \right] d\xi + \int_{-\infty}^0 (-K) \left[\frac{1}{2a\sqrt{\pi t}} e^{-\frac{(x-\xi)^2}{4a^2 t}} \right] d\xi \\ &\quad + \int_{-\infty}^0 (-Ke^{\lambda \xi}) \left[\frac{1}{2a\sqrt{\pi t}} e^{-\frac{(x-\xi)^2}{4a^2 t}} \right] d\xi + \int_0^{\infty} (Ke^{-\lambda \xi}) \left[\frac{1}{2a\sqrt{\pi t}} e^{-\frac{(x-\xi)^2}{4a^2 t}} \right] d\xi \quad (3) \end{aligned}$$

对上式右边前两项积分, 第一项中令 $z = \frac{x-\xi}{2a\sqrt{t}}$, 则第一项积分可化为

$$\int_{-\infty}^{\frac{x}{2a\sqrt{t}}} (-K) \frac{1}{\sqrt{\pi}} e^{-z^2} dz$$

第二项中令 $z = \frac{\xi - x}{2a\sqrt{t}}$, 则第二项积分可化为

$$\int_{-\frac{x}{2a\sqrt{t}}}^{\infty} (-K) \frac{1}{\sqrt{\pi}} e^{-z^2} dz。$$

所以, 前两项积分可合并为

$$\int_{-\frac{x}{2a\sqrt{t}}}^{\frac{x}{2a\sqrt{t}}} (-K) \frac{1}{\sqrt{\pi}} e^{-z^2} dz = (-K) \int_0^{\frac{x}{2a\sqrt{t}}} \frac{2}{\sqrt{\pi}} e^{-z^2} dz = -K \operatorname{erf}\left(\frac{x}{2a\sqrt{t}}\right)$$

对③式右边后两项积分, 对指数部分进行配平方处理, 改写成

$$\int_{-\infty}^0 (-K) \frac{1}{2a\sqrt{\pi t}} e^{-\frac{[\xi - (x+2a^2t\lambda)]^2}{4a^2t}} e^{a^2\lambda^2t} e^{\lambda x} d\xi + \int_0^x (K) \frac{1}{2a\sqrt{\pi t}} e^{-\frac{[\xi - (x-2a^2t\lambda)]^2}{4a^2t}} e^{a^2\lambda^2t} e^{-\lambda x} d\xi$$

第三项中令 $z = \frac{\xi - (x+2a^2\lambda x)}{2a\sqrt{t}}$, 则第三项积分可化为

$$\begin{aligned} \int_{-\infty}^{\frac{x+2a^2t\lambda}{2a\sqrt{t}}} \frac{K}{\sqrt{\pi}} e^{-z^2} e^{a^2\lambda^2t} e^{\lambda x} dz &= \frac{K}{2} e^{a^2\lambda^2t} e^{\lambda x} \left[\int_{-\infty}^0 \frac{2}{\sqrt{\pi}} e^{-z^2} dz + \int_0^{\frac{x+2a^2t\lambda}{2a\sqrt{t}}} \frac{2}{\sqrt{\pi}} e^{-z^2} dz \right] \\ &= \frac{K}{2} e^{a^2\lambda^2t} e^{\lambda x} \left[-1 + \operatorname{erf}\left(\frac{x+2a^2t\lambda}{2a\sqrt{t}}\right) \right] \\ &= -\frac{K}{2} e^{a^2\lambda^2t} e^{\lambda x} \operatorname{erfc}\left(\frac{x+2a^2t\lambda}{2a\sqrt{t}}\right) \end{aligned}$$

第四项中令 $z = \frac{\xi - (x-2a^2\lambda x)}{2a\sqrt{t}}$, 则第四项积分可化为

$$\begin{aligned} \int_{-\frac{x-2a^2t\lambda}{2a\sqrt{t}}}^{\infty} \frac{K}{\sqrt{\pi}} e^{-z^2} e^{a^2\lambda^2t} e^{\lambda x} dz &= \frac{K}{2} e^{a^2\lambda^2t} e^{\lambda x} \left[\int_{-\frac{x-2a^2t\lambda}{2a\sqrt{t}}}^0 \frac{2}{\sqrt{\pi}} e^{-z^2} dz + \int_0^{\infty} \frac{2}{\sqrt{\pi}} e^{-z^2} dz \right] \\ &= \frac{K}{2} e^{a^2\lambda^2t} e^{\lambda x} \left[1 - \operatorname{erf}\left(\frac{-x+2a^2t\lambda}{2a\sqrt{t}}\right) \right] \\ &= \frac{K}{2} e^{a^2\lambda^2t} e^{\lambda x} \operatorname{erfc}\left(\frac{-x+2a^2t\lambda}{2a\sqrt{t}}\right) \end{aligned}$$

结果

$$u(x, t) = \frac{K}{2} e^{a^2\lambda^2t} \left\{ e^{-\lambda x} \operatorname{erfc}\left(\frac{2a^2\lambda t - x}{2a\sqrt{t}}\right) - e^{\lambda x} \operatorname{erfc}\left(\frac{2a^2\lambda t + x}{2a\sqrt{t}}\right) \right\} - K \operatorname{erf}\left(\frac{x}{2a\sqrt{t}}\right)。$$

3. 半无界杆，杆端 $x=0$ 有谐变热流 $B \sin \omega t$ 进入，求长时间以后的杆上温度分布 $u(x, t)$ 。

解：定解问题是

$$\begin{cases} u_t - a^2 u_{xx} = 0, & (x \geq 0) \\ u_x|_{x=0} = -\frac{B}{K} e^{i\omega t}, \end{cases} \quad (1)$$

将边界条件写成 $u_x|_{x=0} = -\frac{B}{K} e^{i\omega t}$ ，在结果中取虚部。根据边界条件，系统长时间后的行为将与边界条件相同的周期变化。将解设为 $u(x, t) = X(x) e^{i\omega t}$ ，以此代入定解问题①，得到

$$\begin{cases} X'' - \frac{i\omega}{a^2} X = 0, \\ X'|_{x=0} = -\frac{B}{K}. \end{cases} \quad (2)$$

②的通解为

$$X = C e^{+\sqrt{\frac{i\omega}{a^2}} x} + D e^{-\sqrt{\frac{i\omega}{a^2}} x},$$

其中 $\sqrt{i} = \sqrt{e^{i\frac{\pi}{2}}} = e^{i\frac{\pi}{4}} = \frac{\sqrt{2}}{2} + i\frac{\sqrt{2}}{2}$ ，从而有

$$X = C e^{+\sqrt{\frac{\omega}{a^2}} \left(\frac{\sqrt{2}}{2} + i\frac{\sqrt{2}}{2} \right) x} + D e^{-\sqrt{\frac{\omega}{a^2}} \left(\frac{\sqrt{2}}{2} + i\frac{\sqrt{2}}{2} \right) x},$$

自然边界条件要求当 $x \rightarrow \infty$ 时， u 有限。置 $C=0$ ，于是有

$$X = D e^{-\sqrt{\frac{\omega}{a^2}} \left(\frac{\sqrt{2}}{2} + i\frac{\sqrt{2}}{2} \right) x}.$$

以此代入边值③以确定系数 D

$$-\sqrt{\frac{\omega}{2a^2}} (1+i) D = -\frac{B}{K},$$

$$D = \frac{aB}{K} \sqrt{\frac{1}{2\omega}} (1-i), \quad (1-i) = \sqrt{2} e^{-i\frac{\pi}{4}}, \Rightarrow D = \frac{aB}{K} \sqrt{\frac{1}{\omega}} e^{-i\frac{\pi}{4}},$$

这样

$$\begin{aligned} u(x, t) &= \frac{aB}{K} \sqrt{\frac{1}{\omega}} e^{-i\frac{\pi}{4}} \cdot e^{-\sqrt{\frac{\omega}{2a^2}} (1+i)x} \cdot e^{i\omega t} \\ &= \frac{aB}{K\sqrt{\omega}} e^{-\sqrt{\frac{\omega}{2a^2}} x} \cdot e^{i\left(\omega t - \sqrt{\frac{\omega}{2a^2}} x - \frac{\pi}{4}\right)}, \end{aligned}$$

取虚部得

$$u(x, t) = \frac{aB}{K\sqrt{\omega}} e^{-\sqrt{\frac{\omega}{2a^2}} x} \sin\left(\omega t - \sqrt{\frac{\omega}{2a^2}} x - \frac{\pi}{4}\right).$$

4. 应用泊松公式计算下述定解问题的解, $u_{tt} - a^2 \Delta u = 0$ 。初始速度为零, 初始位移在某个单位球内为 1, 在球外为零。

解: 以单位球球心为坐标原点, 建立球坐标系, 则定解问题为

$$\begin{cases} u_{tt} - a^2 \Delta u = 0, \\ u|_{t=0} = \begin{cases} 1 & (r < 1) \\ 0 & (r > 1) \end{cases} \\ u_t|_{t=0} = 0. \end{cases}$$

其解由泊松公式

$$u(\mathbf{r}, t) = \frac{1}{4\pi a} \frac{\partial}{\partial t} \iint_{S_{at}^r} \frac{\varphi(\mathbf{r}')}{at} dS' + \frac{1}{4\pi a} \iint_{S_{at}^r} \frac{\psi(\mathbf{r}')}{at} dS'.$$

给出。因初速为零, 则有

$$u(\mathbf{r}, t) = \frac{1}{4\pi a} \frac{\partial}{\partial t} \iint_{S_{at}^r} \frac{\varphi(\mathbf{r}')}{at} dS'.$$

(i) 当点 $\mathbf{r}$ (以 $\mathbf{r}$ 为矢径的点简称为点 $\mathbf{r}$, 下同) 在单位球内

a) 若 $at < 1 - r$, 球面 S_{at}^r 完全位于单位球内 (图 13-1)。从而 $\varphi(\mathbf{r}') = 1$,

$$\begin{aligned} u(\mathbf{r}, t) &= \frac{1}{4\pi a} \frac{\partial}{\partial t} \left(\frac{1}{at} \iint_{S_{at}^r} dS' \right) \\ &= \frac{1}{4\pi a} \frac{\partial}{\partial t} \left[\frac{1}{at} \cdot 4\pi (at)^2 \right] \\ &= 1. \end{aligned}$$

b) 若 $1 - r < at < 1 + r$, 球面 S_{at}^r 与单位球相交 (图 13-2)。设球面 S_{at}^r 在单位球内的部分为 $S_{at,1}^r$ 。在 $S_{at,1}^r$ 上, $\varphi(\mathbf{r}') = 1$, 而在球外部分 $\varphi(\mathbf{r}') = 0$,

$$u(\mathbf{r}, t) = \frac{1}{4\pi a} \frac{\partial}{\partial t} \iint_{S_{at,1}^r} \frac{1}{at} dS',$$

其中,

$$\begin{aligned} \iint_{S_{at,1}^r} dS' &= \int_0^{2\pi} \int_0^{\theta_0} (at)^2 \sin \theta d\theta d\varphi \\ &= 2\pi (at)^2 (-\cos \theta) \Big|_0^{\theta_0} \\ &= 2\pi (at)^2 (1 - \cos \theta_0) \\ &= 2\pi (at)^2 \left(1 - \frac{r^2 + a^2 t^2 - 1}{2rat} \right) \\ &= \frac{\pi at}{r} [1 - (r - at)^2], \end{aligned}$$

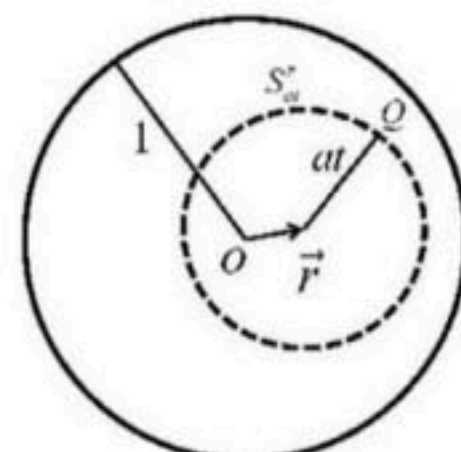


图13-1

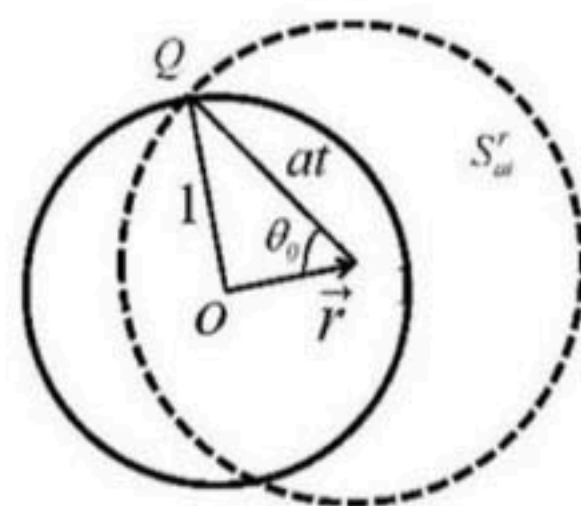


图13-2

$$u(\mathbf{r}, t) = \frac{1}{4\pi a} \frac{\partial}{\partial t} \left\{ \frac{\pi}{r} [1 - (r - at)^2] \right\}$$

$$= \frac{1}{2r} (r - at)。$$

c) 若 $at > r + 1$, 球面 S_{at}^r 完全包含单位球面 (图 13-3)。从而 $\varphi(\mathbf{r}') = 0$, 有

$$u(\mathbf{r}, t) = 0。$$

(ii) 当点 $\frac{1}{r}$ 在单位球外时:

a) 若 $at + 1 < r$, 球面 S_{at}^r 与单位球分离。在 S_{at}^r 上 $\varphi(\mathbf{r}') = 0$,

$$u(\mathbf{r}, t) = 0。$$

b) 若 $at > r + 1$, 球面 S_{at}^r 完全包含单位球面。在 S_{at}^r 上 $\varphi(\mathbf{r}') = 0$, 从而有

$$u(\mathbf{r}, t) = 0。$$

c) 若 $r - 1 < at < 1 + r$, 球面 S_{at}^r 与单位球相交 (图 13-4)。设球面 S_{at}^r 在单位球内的部分为 $S_{at,1}^r$ 。在 $S_{at,1}^r$ 上, 与(i)之 c) 计算相同, 得

$$u(\mathbf{r}, t) = \frac{1}{2r} (r - at)。$$

综合结果, 在球内:

$$u(\mathbf{r}, t) = \begin{cases} 1, & \left(t < \frac{1-r}{a} \right) \\ 0, & \left(t > \frac{1+r}{a} \right) \\ \frac{1}{2r} (r - at), & \left(\frac{1-r}{a} < t < \frac{1+r}{a} \right) \end{cases}$$

在球外:

$$u(\mathbf{r}, t) = \begin{cases} 0, & \left(t < \frac{r-1}{a} \right) \\ 0, & \left(t > \frac{r+1}{a} \right) \\ u(\mathbf{r}, t) = \frac{1}{2r} (r - at), & \left(\frac{r-1}{a} < t < \frac{r+1}{a} \right) \end{cases}$$

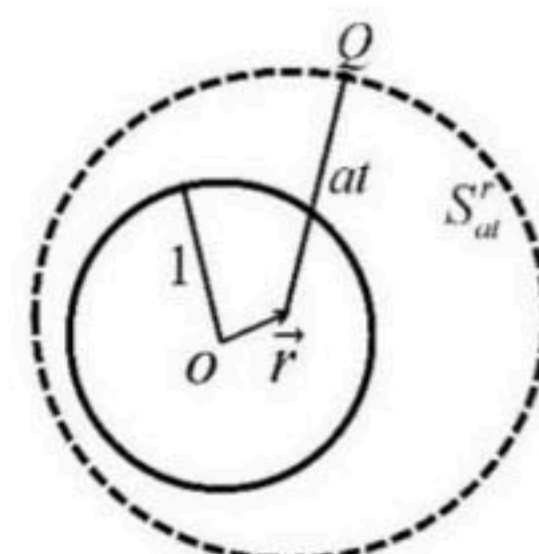


图13-3

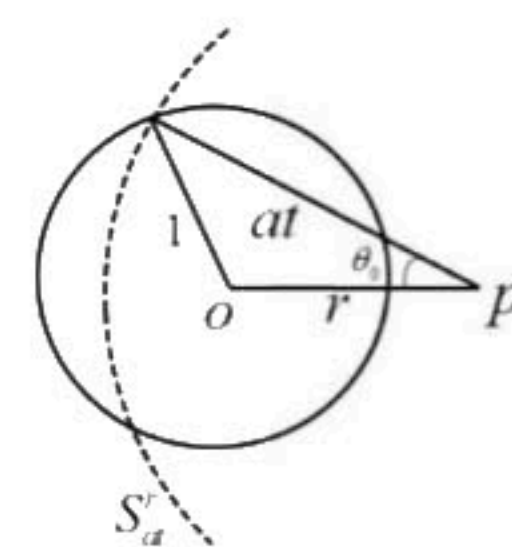


图13-4

5. 应用泊松公式计算下述定解问题的解。 $u_{tt} - a^2 \Delta u = 0$, 初始速度为零, 初始位移在球 $r = r_0$ 以内为 $A \cos(\pi r / 2r_0)$, 在球外为零。

解: 以单位球球心为坐标原点, 建立球坐标系, 则定解问题为

$$\begin{cases} u_{tt} - a^2 \Delta u = 0, \\ u|_{t=0} = \begin{cases} A \cos(\pi r / 2r_0), & (r < r_0) \\ 0, & (r > r_0) \end{cases} \\ u_t|_{t=0} = 0. \end{cases}$$

其解由泊松公式

$$u(\mathbf{r}, t) = \frac{1}{4\pi a} \frac{\partial}{\partial t} \iint_{S_{at}^r} \frac{\varphi(\mathbf{r}')}{at} dS' + \frac{1}{4\pi a} \iint_{S_{at}^r} \frac{\psi(\mathbf{r}')}{at} dS',$$

给出。因初速为零, 则有

$$u(\mathbf{r}, t) = \frac{1}{4\pi a} \frac{\partial}{\partial t} \iint_{S_{at}^r} \frac{\varphi(\mathbf{r}')}{at} dS'.$$

(i) 当点 $\mathbf{r}$ (以 $\mathbf{r}$ 为径矢的点简称为点 $\mathbf{r}$, 下同) 在 r_0 球内

a) 当 $at < r_0 - r$, 球面 S_{at}^r 完全位于 r_0 球内 (图 13-5)。从而

$$u(\mathbf{r}, t) = \frac{1}{4\pi a} \frac{\partial}{\partial t} \iint_{S_{at}^r} \frac{A \cos \frac{\pi r'}{2r_0}}{at} dS' = \frac{1}{4\pi a} \frac{\partial}{\partial t} \int_0^\pi \frac{A \cos \frac{\pi r'}{2r_0}}{at} \cdot 2\pi(at)^2 \sin \theta d\theta.$$

为计算上式右端积分, 参见图 13-5, 由余弦定理, 有 $r'^2 = r^2 + (at)^2 - 2rat \cos \theta$ 。由此有 $2r'dr' = 2r \cdot at \cdot \sin \theta d\theta$,

$$\begin{aligned} u(\mathbf{r}, t) &= \frac{1}{4\pi a} \frac{\partial}{\partial t} \int_{|r-at|}^{r+at} \frac{A \cos \frac{\pi r'}{2r_0}}{at} \cdot 2\pi(at)^2 \frac{r'}{rat} dr' \\ &= \frac{A}{2ra} \frac{\partial}{\partial t} \int_{|r-at|}^{r+at} r' \cos \frac{\pi r'}{2r_0} dr'. \end{aligned}$$

利用含参变量积分求导公式, 有

$$\begin{aligned} \frac{\partial}{\partial t} \int_{r-at}^{r+at} f(r') dr' &= af(r+at) - (-a)f(r-at) \\ &= a[f(r+at) + f(r-at)], \\ \frac{\partial}{\partial t} \int_{at-r}^{r+at} f(r') dr' &= af(r+at) - af(at-r) \\ &= a[f(r+at) - f(at-r)]. \end{aligned}$$

从而

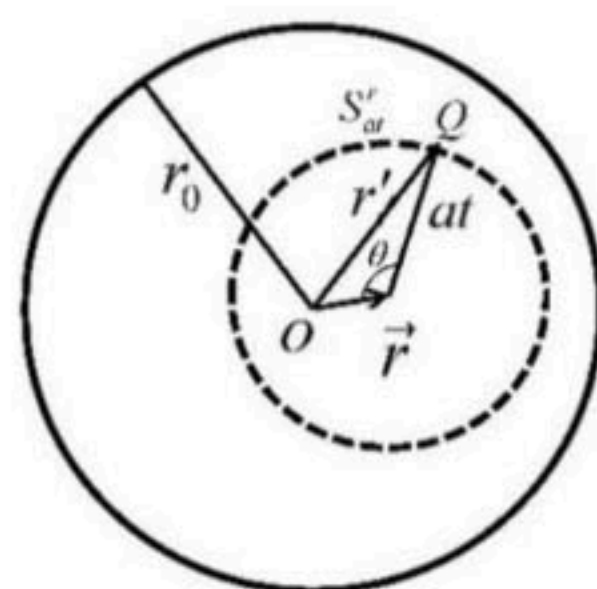


图13-5

$$u(\mathbf{r}, t) = \begin{cases} \frac{A}{2r} \left[(r+at) \cos \frac{\pi(r+at)}{2r_0} + (r-at) \cos \frac{\pi(r-at)}{2r_0} \right], & (at < r) \\ \frac{A}{2r} \left[(r+at) \cos \frac{\pi(r+at)}{2r_0} + (at-r) \cos \frac{\pi(at-r)}{2r_0} \right], & (at > r) \end{cases}$$

$$= \frac{A}{2r} \left[(r+at) \cos \frac{\pi(r+at)}{2r_0} + (r-at) \cos \frac{\pi(r-at)}{2r_0} \right].$$

b) 当 $r_0 - r < at < r_0 + r$, 球面 S_{at}^r 与单位球相交 (图 13-6)。计算与 a) 类似, 只是这里只需要计算 S_{at}^r 在单位球内的部分。

$$u(\mathbf{r}, t) = \frac{1}{4\pi a} \frac{\partial}{\partial t} \iint_{S_{at}^r} \frac{A \cos \frac{\pi r'}{2r_0}}{at} dS'$$

$$= \frac{1}{4\pi a} \frac{\partial}{\partial t} \int_0^{\theta_0} \frac{A \cos \frac{\pi r'}{2r_0}}{at} \cdot 2\pi (at)^2 \sin \theta d\theta$$

为计算上式右端积分, 参见图 13-6, 由余弦定理, 有, $r'^2 = r^2 + (at)^2 - 2rat \cos \theta$ 。由此有 $2r' dr' = 2r \cdot at \cdot \sin \theta d\theta$,

$$u(\mathbf{r}, t) = \frac{1}{4\pi a} \frac{\partial}{\partial t} \int_{|r-at|}^{r_0} \frac{A \cos \frac{\pi r'}{2r_0}}{at} \cdot 2\pi (at)^2 \frac{r'}{rat} dr'$$

$$= \frac{A}{2r} (r-at) \cos \frac{\pi(r-at)}{2r_0}.$$

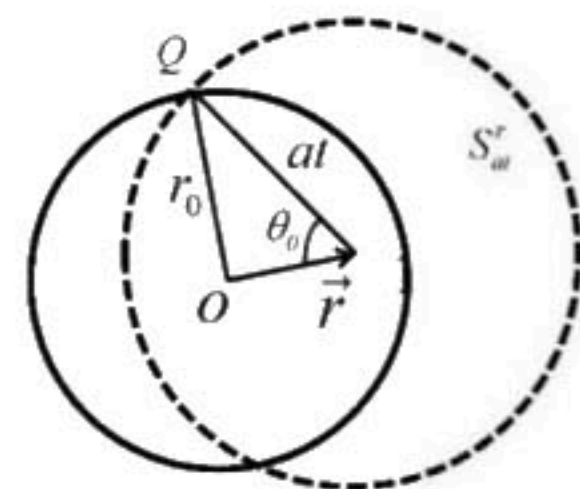


图13-6

c) 当 $at > r + r_0$, 球面 S_{at}^r 完全包含单位球面。从而 $\varphi(\mathbf{r}') = 0$, 有 $u(\mathbf{r}, t) = 0$ 。

(ii) 当点 $\frac{1}{r}$ 在单位球外时:

a) 当 $at < r - r_0$, 球面 S_{at}^r 与单位球分离。在 S_{at}^r 上 $\varphi(\mathbf{r}') = 0$, $u(\mathbf{r}, t) = 0$ 。

b) 当 $r - r_0 < at < r + r_0$, 球面 S_{at}^r 与单位球相交 (图 13-7)。设球面 S_{at}^r 在单位球内的部分为 $S_{at,1}^r$ 。在 $S_{at,1}^r$ 上, 与(i)之 c) 计算相同, 得

$$u(\mathbf{r}, t) = \frac{A}{2r} (r-at) \cos \frac{\pi(r-at)}{2r_0}.$$

c) 当 $at > r + r_0$, 球面 S_{at}^r 完全包含单位球面。在 S_{at}^r 上 $\varphi(\mathbf{r}') = 0$, 从而有

$$u(\mathbf{r}, t) = 0.$$

综合结果, 在球内:

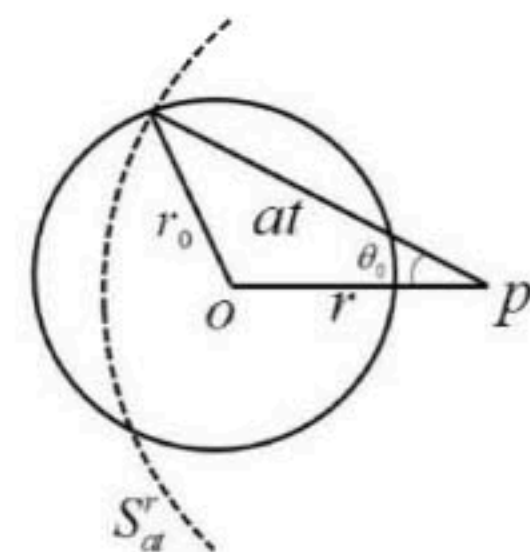


图13-7

$$u(\mathbf{r}, t) = \begin{cases} \frac{A}{2r} \left[(r+at) \cos \frac{\pi(r+at)}{2r_0} + (r-at) \cos \frac{\pi(r-at)}{2r_0} \right], & (at < r_0 - r) \\ \frac{A}{2r} (r-at) \cos \frac{\pi(r-at)}{2r_0}, & (r_0 - r < at < r_0 + r) \\ 0, & (at > r_0 + r) \end{cases}$$

在球外:

$$u(\mathbf{r}, t) = \begin{cases} 0, & (at < r - r_0) \\ \frac{A}{2r} (r-at) \cos \frac{\pi(r-at)}{2r_0}, & (r - r_0 < at < r + r_0) \\ 0, & (at > r + r_0) \end{cases}$$

6. 二维波动, 初始速度为零, 初始位移在圆 $\rho=1$ 以内为 1, 在圆外为零. 试求 $u|_{\rho=0}$.

解: 以圆柱轴心为坐标原点, 建立柱坐标系, 则定解问题为

$$\begin{cases} u_{tt} - a^2 \Delta u = 0, \\ u|_{t=0} = \begin{cases} 1, & (\rho < 1) \\ 0, & (\rho > 1) \end{cases} \\ u_t|_{t=0} = 0. \end{cases}$$

其解由二维泊松公式

$$u(x, y, t) = \frac{1}{2\pi a} \frac{\partial}{\partial t} \iint_{\Sigma_{at}^{x,y}} \frac{\varphi(x', y')}{\sqrt{a^2 t^2 - (x' - x)^2 - (y' - y)^2}} dx' dy' \\ + \frac{1}{2\pi a} \iint_{\Sigma_{at}^{x,y}} \frac{\psi(x', y')}{\sqrt{a^2 t^2 - (x' - x)^2 - (y' - y)^2}} dx' dy'$$

给出. 由定解条件知 $\psi = 0$, 则

$$u(0, 0, t) = \frac{1}{2\pi a} \frac{\partial}{\partial t} \iint_{\Sigma_{at}^{x,y}} \frac{\varphi(x', y')}{\sqrt{a^2 t^2 - x'^2 - y'^2}} dx' dy'$$

换到极坐标

$$u(0, t) = \frac{1}{2\pi a} \frac{\partial}{\partial t} \int_0^{2\pi} d\theta' \int_0^1 \frac{\rho' d\rho'}{\sqrt{a^2 t^2 - \rho'^2}}.$$

(i) 当 $at < 1$

$$\begin{aligned} u(0, t) &= \frac{1}{2\pi a} \frac{\partial}{\partial t} \int_0^{2\pi} d\theta' \int_0^{at} \frac{\rho' d\rho'}{\sqrt{a^2 t^2 - \rho'^2}} \\ &= \frac{1}{2\pi a} \cdot 2\pi \cdot \frac{\partial}{\partial t} \left[\frac{1}{2} \left(-2\sqrt{a^2 t^2 - \rho'^2} \right) \Big|_0^{at} \right] \\ &= \frac{1}{a} \frac{\partial}{\partial t} (at) \\ &= 1. \end{aligned}$$

(ii) 当 $at > 1$

$$\begin{aligned} u(0, t) &= \frac{1}{2\pi a} \frac{\partial}{\partial t} \int_0^{2\pi} d\theta' \int_0^1 \frac{\rho' d\rho'}{\sqrt{a^2 t^2 - \rho'^2}} \\ &= \frac{1}{2\pi a} \cdot 2\pi \cdot \frac{\partial}{\partial t} \left[\frac{1}{2} \left(-2\sqrt{a^2 t^2 - \rho'^2} \right) \Big|_0^1 \right] \\ &= \frac{1}{a} \frac{\partial}{\partial t} \left(at - \sqrt{a^2 t^2 - 1} \right) \\ &= \frac{1}{a} \frac{\partial}{\partial t} \left(at - \sqrt{a^2 t^2 - 1} \right) \\ &= 1 - \frac{at}{\sqrt{a^2 t^2 - 1}}. \end{aligned}$$

7. 求解三维无界空间中的输运问题 $u_t - a^2 \Delta u = 0, u|_{t=0} = \varphi(x, y, z)$ 。

解：定解问题

$$\begin{cases} u_t - a^2 \Delta u = 0, \\ u|_{t=0} = \varphi(x, y, z). \end{cases}$$

作傅里叶变换，定解问题变换为

$$\begin{cases} U' + k^2 a^2 U = 0, \\ U|_{t=0} = \Phi(k_1, k_2, k_3). \end{cases}$$

这个常微分方程的初始值问题的解是

$$U(t, \vec{k}) = \Phi(\vec{k}) e^{-k^2 a^2 t}.$$

再进行逆傅里叶变换，

$$\begin{aligned} u(\vec{r}, t) &= \mathcal{F}^{-1}[U(t, \vec{k})] = \int_{-\infty}^{\infty} \Phi(\vec{k}) e^{-k^2 a^2 t} e^{i\vec{k} \cdot \vec{r}} d\vec{k} \\ &= \frac{1}{2\pi} \int_{-\infty}^{\infty} \left[\int_{-\infty}^{\infty} \varphi(\vec{r}') e^{-i\vec{k} \cdot \vec{r}'} d\vec{r}' \right] e^{-k^2 a^2 t} e^{i\vec{k} \cdot \vec{r}} d\vec{k} \end{aligned}$$

交换积分次序

$$u(\vec{r}, t) = \frac{1}{2\pi} \int_{-\infty}^{\infty} \varphi(\vec{r}') \left[\int_{-\infty}^{\infty} e^{-k^2 a^2 t} e^{i\vec{k} \cdot (\vec{r} - \vec{r}')} d\vec{k} \right] d\vec{r}'.$$

引用教材本节例 2 的积分公式 $\int_{-\infty}^{\infty} e^{-\alpha^2 k^2} e^{\beta k} dk = (\sqrt{\pi}/\alpha) e^{\beta^2/4\alpha^2}$ 计算 [...] 内的积分，置

$\alpha = a\sqrt{t}, \beta = i(x - \xi)$ 以利用此积分公式，即得

$$u(\vec{r}, t) = \int_{-\infty}^{\infty} \varphi(\vec{r}') \left[\frac{1}{(2a\sqrt{\pi t})^3} e^{-\frac{|\vec{r} - \vec{r}'|^2}{4a^2 t}} \right] d\vec{r}'.$$

8. 例 6 研究三维无界空间中的自由振动是从初始 ($t=0$) 状态推算以后 ($t>0$) 的状态。试重新求解例 6, 从初始状态反推以前 ($t<0$) 的状态。

解: 定解问题是

$$\begin{cases} u_{tt} - a^2 \Delta_3 u = 0, \\ u|_{t=0} = \varphi(\mathbf{r}), \quad U_t|_{t=0} = \psi(\mathbf{r}). \end{cases} \quad (1)$$

对①作傅里叶变换, 问题变换为常微分方程的初始值问题

$$\begin{cases} U'' + k^2 a^2 U = 0, \\ U|_{t=0} = \Phi(\mathbf{k}), U'|_{t=0} = \Psi(\mathbf{k}). \end{cases}$$

这个问题的解是

$$U(t, \mathbf{k}) = \frac{1}{2} \Phi(\mathbf{k})(e^{ikat} + e^{-ikat}) + \frac{1}{2a} \frac{1}{ik} \Psi(\mathbf{k})(e^{ikat} - e^{-ikat}).$$

再进行逆傅里叶变换,

$$\begin{aligned} u(\mathbf{r}, t) &= \iiint \left[\Phi(\mathbf{k}) \frac{1}{2} (e^{ikat} + e^{-ikat}) + \Psi(\mathbf{k}) \frac{1}{2a} \frac{1}{ik} (e^{ikat} - e^{-ikat}) \right] e^{i\mathbf{k} \cdot \mathbf{r}} d\mathbf{k}_1 d\mathbf{k}_2 d\mathbf{k}_3 \\ &= \frac{1}{4\pi a} \iiint_{-\infty}^{\infty} \varphi(\mathbf{r}') \left[\iiint_{-\infty}^{\infty} \frac{a}{4\pi^2} (e^{ikat} + e^{-ikat}) e^{i\mathbf{k} \cdot (\mathbf{r} - \mathbf{r}')} d\mathbf{k}_1 d\mathbf{k}_2 d\mathbf{k}_3 \right] dV' \\ &\quad + \frac{1}{4\pi a} \iiint_{-\infty}^{\infty} \psi(\mathbf{r}') \left[\iiint_{-\infty}^{\infty} \frac{1}{4\pi^2} \frac{1}{ik} (e^{ikat} - e^{-ikat}) e^{i\mathbf{k} \cdot (\mathbf{r} - \mathbf{r}')} d\mathbf{k}_1 d\mathbf{k}_2 d\mathbf{k}_3 \right] dV' \\ &= \frac{1}{4\pi a} \frac{\partial}{\partial t} \iiint_{-\infty}^{\infty} \varphi(\mathbf{r}') \left[\iiint_{-\infty}^{\infty} \frac{1}{4\pi^2} \frac{1}{ik} (e^{ikat} - e^{-ikat}) e^{i\mathbf{k} \cdot (\mathbf{r} - \mathbf{r}')} d\mathbf{k}_1 d\mathbf{k}_2 d\mathbf{k}_3 \right] dV' \\ &\quad + \frac{1}{4\pi a} \iiint_{-\infty}^{\infty} \psi(\mathbf{r}') \left[\iiint_{-\infty}^{\infty} \frac{1}{4\pi^2} \frac{1}{ik} (e^{ikat} - e^{-ikat}) e^{i\mathbf{k} \cdot (\mathbf{r} - \mathbf{r}')} d\mathbf{k}_1 d\mathbf{k}_2 d\mathbf{k}_3 \right] dV'. \end{aligned}$$

引用 § 5.3 例 1 结果,

$$\begin{aligned} u(\mathbf{r}, t) &= \frac{1}{4\pi a} \frac{\partial}{\partial t} \iiint_{-\infty}^{\infty} \varphi(\mathbf{r}') \\ &\quad \times \iiint_{-\infty}^{\infty} \mathcal{S} \left\{ \frac{1}{r} [\delta(r-at) + \delta(r+at)] e^{-ik \cdot \mathbf{r}'} \right\} e^{i\mathbf{k} \cdot \mathbf{r}} d\mathbf{k}_1 d\mathbf{k}_2 d\mathbf{k}_3 dV' \\ &\quad + \frac{1}{4\pi a} \iiint_{-\infty}^{\infty} \psi(\mathbf{r}') \\ &\quad \times \iiint_{-\infty}^{\infty} \mathcal{S} \left\{ \frac{1}{r} [\delta(r-at) + \delta(r+at)] e^{-ik \cdot \mathbf{r}'} \right\} e^{i\mathbf{k} \cdot \mathbf{r}} d\mathbf{k}_1 d\mathbf{k}_2 d\mathbf{k}_3 dV'. \end{aligned}$$

应用延迟定理,

$$\begin{aligned} u(\mathbf{r}, t) &= \frac{1}{4\pi a} \frac{\partial}{\partial t} \iiint_{-\infty}^{\infty} \frac{\varphi(\mathbf{r}')}{|\mathbf{r} - \mathbf{r}'|} [\delta(|\mathbf{r} - \mathbf{r}'| - at) + \delta(|\mathbf{r} - \mathbf{r}'| + at)] dV' \\ &\quad + \frac{1}{4\pi a} \iiint_{-\infty}^{\infty} \frac{\psi(\mathbf{r}')}{|\mathbf{r} - \mathbf{r}'|} [\delta(|\mathbf{r} - \mathbf{r}'| - at) + \delta(|\mathbf{r} - \mathbf{r}'| + at)] dV' \end{aligned}$$

其中 $\delta(|\mathbf{r} - \mathbf{r}'| - at)$ 对应 $t>0$, 表示是从初始 ($t=0$) 状态推算以后 ($t>0$) 的状态。
 $\delta(|\mathbf{r} - \mathbf{r}'| + at)$ 对应 $t<0$, 表示从初始状态反推以前 ($t<0$) 的状态。合起来, 可写为

$$\begin{aligned} u(\mathbf{r}, t) &= \frac{1}{4\pi a} \frac{\partial}{\partial t} \iiint_{-\infty}^{\infty} \frac{\varphi(\mathbf{r}')}{|\mathbf{r} - \mathbf{r}'|} \delta(|\mathbf{r} - \mathbf{r}'| - a|t|) dV' \\ &\quad + \frac{1}{4\pi a} \iiint_{-\infty}^{\infty} \frac{\psi(\mathbf{r}')}{|\mathbf{r} - \mathbf{r}'|} \delta(|\mathbf{r} - \mathbf{r}'| - a|t|) dV' \end{aligned}$$

由于被积式中出现 $\delta(|\mathbf{r} - \mathbf{r}'| - a|t|)$, 对 $\mathbf{r}'$ 的积分只需在球面 $S_{a|t|}^{\mathbf{r}}$ 上进行, $S_{a|t|}^{\mathbf{r}}$ 以点 $\mathbf{r}$ (确切地说, 径矢为 $\mathbf{r}$ 的点) 为球心, 而半径为 at ,

$$u(\mathbf{r}, t) = \frac{1}{4\pi a} \frac{\partial}{\partial t} \iint_{S_{a|t|}^{\mathbf{r}}} \frac{\varphi(\mathbf{r}')}{at} dS' + \frac{1}{4\pi a} \iint_{S_{a|t|}^{\mathbf{r}}} \frac{\psi(\mathbf{r}')}{at} dS', \quad (2)$$

式中 dS' 是球面 $S_{a|t|}^{\mathbf{r}}$ 的面积元。② 式称为泊松公式。

9. 求解一维半无界空间的输运问题 $u_t - a^2 u_{xx} = 0$, $u|_{x=0} = At$, $u|_{t=0} = 0$ 。

解：定解问题

$$\begin{cases} u_t - a^2 u_{xx} = 0, & (x > 0) \\ u|_{x=0} = At, \\ u|_{t=0} = 0. \end{cases} \quad (1)$$

令 $u = w + At$, 定解问题①转化成 w 的定解问题

$$\begin{cases} w_t - a^2 w_{xx} = -A, & (x > 0) \\ w|_{x=0} = 0, \\ w|_{t=0} = 0. \end{cases}$$

将问题延拓到 $x < 0$ 的半空间

$$\begin{cases} w_t - a^2 w_{xx} = \begin{cases} -A & x > 0 \\ A & x < 0 \end{cases} \\ w|_{t=0} = 0. \end{cases}$$

本题解答由(13.1.3)给出

$$\begin{aligned} w(x, t) &= \int_0^t \int_{-\infty}^{\infty} f(\xi, \tau) \left[\frac{1}{2a\sqrt{\pi(t-\tau)}} e^{-\frac{(x-\xi)^2}{4a^2(t-\tau)}} \right] d\xi d\tau \\ &= \int_0^t \int_{-\infty}^0 A \left[\frac{1}{2a\sqrt{\pi(t-\tau)}} e^{-\frac{(x-\xi)^2}{4a^2(t-\tau)}} \right] d\xi d\tau \\ &\quad + \int_0^t \int_0^{\infty} (-A) \left[\frac{1}{2a\sqrt{\pi(t-\tau)}} e^{-\frac{(x-\xi)^2}{4a^2(t-\tau)}} \right] d\xi d\tau, \end{aligned} \quad (2)$$

在②右边第一个积分中令 $z = (x - \xi) / 2a\sqrt{t - \tau}$, ②右边第二个积分中令 $z = (\xi - x) / 2a\sqrt{t - \tau}$, 则有

$$\begin{aligned} w(x, t) &= -\int_{\tau=0}^t \left(\frac{A}{\sqrt{\pi}} \int_{\frac{x}{2a\sqrt{t-\tau}}}^{\infty} e^{-z^2} dz + \frac{A}{\sqrt{\pi}} \int_{-\infty}^{\frac{-x}{2a\sqrt{t-\tau}}} e^{-z^2} dz \right) d\tau \\ &= -A \int_{\tau=0}^t \left(\frac{2}{\sqrt{\pi}} \int_0^{\frac{x}{2a\sqrt{t-\tau}}} e^{-z^2} dz \right) d\tau \\ &= -A \int_{\tau=0}^t \operatorname{erf} \left(\frac{x}{2a\sqrt{t-\tau}} \right) d\tau \end{aligned}$$

从而

$$\begin{aligned} u(x, t) &= At - A \int_0^t \operatorname{erf} \left(\frac{x}{2a\sqrt{t-\tau}} \right) d\tau \\ &= A \int_0^t \left[1 - \operatorname{erf} \left(\frac{x}{2a\sqrt{t-\tau}} \right) \right] d\tau \\ &= A \int_0^t \operatorname{erfc} \left(\frac{x}{2a\sqrt{t-\tau}} \right) d\tau. \end{aligned}$$

10. 在一维半无界空间中求解 $u_t - a^2 u_{xx} = 0, u|_{x=0} = f(t), u|_{t=0} = \varphi(x)$.

解：定解问题

$$\begin{cases} u_t - a^2 u_{xx} = 0, & (x > 0) \\ u|_{x=0} = f(t), \\ u|_{t=0} = \varphi(x). \end{cases} \quad (1)$$

令 $u = w + f(t)$, 定解问题①转化成 w 的定解问题

$$\begin{cases} w_t - a^2 w_{xx} = -f'(t), & (x > 0) \\ w|_{x=0} = 0, \\ w|_{t=0} = \varphi(x) - f(0). \end{cases} \quad (2)$$

先将问题奇延拓到 $x < 0$ 的半空间

$$\begin{cases} w_t - a^2 w_{xx} = f_0(t) = \begin{cases} -f'(t), & (x > 0) \\ f'(t), & (x < 0) \end{cases} \\ w|_{t=0} = \varphi_0(x) = \begin{cases} \varphi(x) - f(0), & (x > 0) \\ -\varphi(-x) + f(0), & (x < 0) \end{cases} \end{cases} \quad (3)$$

对③作傅里叶变换，问题转化成

$$\begin{cases} W' + k^2 a^2 W = F_0(t, k) \\ W|_{t=0} = \Phi_0(k) \end{cases} \quad (4)$$

为求解这个非齐次常微分方程，用 $e^{k^2 a^2 t}$ 遍乘方程各项，得

$$\frac{d}{dt} [W(t; k) e^{k^2 a^2 t}] = F_0(t, k) e^{k^2 a^2 t}.$$

对 t 积分一次，计及始值，

$$W(t; k) e^{k^2 a^2 t} - \Phi_0(k) = \int_0^t F_0(\tau) e^{k^2 a^2 \tau} d\tau$$

即

$$\begin{aligned} W(t; k) &= \Phi_0(k) e^{-k^2 a^2 t} + e^{-k^2 a^2 t} \int_0^t F_0(\tau, k) e^{k^2 a^2 \tau} d\tau \\ &= \frac{1}{2\pi} \int_{-\infty}^{\infty} \varphi_0(\xi) e^{-ik\xi} d\xi e^{-k^2 a^2 t} + \frac{1}{2\pi} \int_0^t \int_{-\infty}^{\infty} f_0(\xi, \tau) e^{-ik\xi} d\xi e^{-k^2 a^2 (t-\tau)} d\tau \end{aligned}$$

进行逆傅里叶变换，

$$\begin{aligned} w(x, t) &= \frac{1}{2\pi} \int_{-\infty}^{\infty} \left[\int_{-\infty}^{\infty} \varphi_0(\xi) e^{-ik\xi} d\xi e^{-k^2 a^2 t} \right] e^{ikx} dk \\ &\quad + \frac{1}{2\pi} \int_{-\infty}^{\infty} \left[\int_0^t \int_{-\infty}^{\infty} f_0(\xi, \tau) e^{-ik\xi} d\xi e^{-k^2 a^2 (t-\tau)} d\tau \right] e^{ikx} dk. \end{aligned}$$

交换积分次序

$$w(x, t) = \frac{1}{2\pi} \int_{-\infty}^{\infty} \varphi_0(\xi) \left[\int_{-\infty}^{\infty} e^{-k^2 a^2 t} e^{ik(x-\xi)} dk \right] d\xi \\ + \frac{1}{2\pi} \int_0^t \int_{-\infty}^{\infty} f_0(\xi, \tau) \left[\int_{-\infty}^{\infty} e^{-k^2 a^2 (t-\tau)} e^{ik(x-\xi)} dk \right] d\xi d\tau.$$

引用教材本节例 2 的积分公式 $\int_{-\infty}^{\infty} e^{-\alpha^2 k^2} e^{\beta k} dk = (\sqrt{\pi}/\alpha) e^{\beta^2/4\alpha^2}$ 计算[]内的积分,

置 $\alpha = a\sqrt{t}, \beta = i(x-\xi)$ 以利用此积分公式, 即得

$$w(x, t) = \int_{-\infty}^{\infty} \varphi_0(\xi) \left[\frac{1}{2a\sqrt{\pi t}} e^{-\frac{(x-\xi)^2}{4a^2 t}} \right] d\xi + \int_0^t \int_{-\infty}^{\infty} f_0(\xi, \tau) \left[\frac{1}{2a\sqrt{\pi(t-\tau)}} e^{-\frac{(x-\xi)^2}{4a^2(t-\tau)}} \right] d\xi d\tau.$$

代入 $f_0(x)$ 和 $\varphi_0(t)$ 的具体表达式, 得到

$$w(x, t) = \int_{-\infty}^0 [-\varphi(-\xi) + f(0)] \left[\frac{1}{2a\sqrt{\pi t}} e^{-\frac{(x-\xi)^2}{4a^2 t}} \right] d\xi + \int_0^{\infty} [\varphi(\xi) - f(0)] \left[\frac{1}{2a\sqrt{\pi t}} e^{-\frac{(x-\xi)^2}{4a^2 t}} \right] d\xi \\ + \int_0^t \int_{-\infty}^0 f'(\tau) \left[\frac{1}{2a\sqrt{\pi(t-\tau)}} e^{-\frac{(x-\xi)^2}{4a^2(t-\tau)}} \right] d\xi d\tau \\ + \int_0^t \int_0^{\infty} (-f'(\tau)) \left[\frac{1}{2a\sqrt{\pi(t-\tau)}} e^{-\frac{(x-\xi)^2}{4a^2(t-\tau)}} \right] d\xi d\tau, \quad (5)$$

在⑤式右边第一个积分中令 $z = (x-\xi)/2a\sqrt{t}$, 则第一个积分可化为

$$-\int_{\infty}^{x/2a\sqrt{t}} [-\varphi(-\xi + 2a\sqrt{t}z) + f(0)] \left[\frac{1}{\sqrt{\pi}} e^{-z^2} \right] dz.$$

在⑤式右边第二个积分中令 $z = (\xi-x)/2a\sqrt{t}$, 则第二个积分可化为

$$\int_{-x/2a\sqrt{t}}^{\infty} [\varphi(\xi + 2a\sqrt{t}z) - f(0)] \left[\frac{1}{\sqrt{\pi}} e^{-z^2} \right] dz.$$

这样, ⑤式前两项积分合并为

$$-f(0) \left[\int_{\infty}^{x/2a\sqrt{t}} \frac{1}{\sqrt{\pi}} e^{-z^2} dz + \int_{-x/2a\sqrt{t}}^{\infty} \frac{1}{\sqrt{\pi}} e^{-z^2} dz \right] \\ + \int_{-x/2a\sqrt{t}}^{\infty} \varphi(x + 2a\sqrt{t}z) \frac{1}{\sqrt{\pi}} e^{-z^2} dz - \int_{x/2a\sqrt{t}}^{\infty} \varphi(-x + 2a\sqrt{t}z) \frac{1}{\sqrt{\pi}} e^{-z^2} dz$$

进一步合并得

$$-f(0)\text{erf}(x/2a\sqrt{t}) \\ + \int_{-x/2a\sqrt{t}}^{\infty} \varphi(x + 2a\sqrt{t}z) \frac{1}{\sqrt{\pi}} e^{-z^2} dz - \int_{x/2a\sqrt{t}}^{\infty} \varphi(-x + 2a\sqrt{t}z) \frac{1}{\sqrt{\pi}} e^{-z^2} dz$$

在⑤式右边第三个积分中令 $z = (x-\xi)/2a\sqrt{t-\tau}$, 则第三个积分可化为

$$-\int_0^t \int_{\infty}^{x/2a\sqrt{t-\tau}} f'(\tau) \left[\frac{1}{\sqrt{\pi}} e^{-z^2} \right] dz d\tau,$$

在⑤式右边第四个积分中令 $z = (\xi - x) / 2a\sqrt{t - \tau}$, 则第四个积分可化为

$$-\int_0^t \int_{-x/2a\sqrt{t-\tau}}^{\infty} f'(\tau) \left[\frac{1}{\sqrt{\pi}} e^{-z^2} \right] dz d\tau.$$

这样, ⑤式后两项积分合并为

$$-\int_0^t f'(\tau) \int_{-x/2a\sqrt{t-\tau}}^{x/2a\sqrt{t-\tau}} \left[\frac{1}{\sqrt{\pi}} e^{-z^2} \right] dz d\tau = -\int_0^t f'(\tau) \operatorname{erf} \left(\frac{x}{2a\sqrt{t-\tau}} \right) d\tau$$

结果是

$$\begin{aligned} w(x, t) = & -\int_0^t f'(\tau) \operatorname{erf} \left(\frac{x}{2a\sqrt{t-\tau}} \right) d\tau - f(0) \operatorname{erf}(x / 2a\sqrt{t}) \\ & + \int_{-x/2a\sqrt{t}}^{\infty} \varphi(x + 2a\sqrt{t}z) \frac{1}{\sqrt{\pi}} e^{-z^2} dz - \int_{x/2a\sqrt{t}}^{\infty} \varphi(-x + 2a\sqrt{t}z) \frac{1}{\sqrt{\pi}} e^{-z^2} dz \end{aligned}$$

从而

$$\begin{aligned} u(x, t) = & f(t) - \int_0^t f'(\tau) \operatorname{erf} \left(\frac{x}{2a\sqrt{t-\tau}} \right) d\tau - f(0) \operatorname{erf}(x / 2a\sqrt{t}) \\ & + \int_{-x/2a\sqrt{t}}^{\infty} \varphi(x + 2a\sqrt{t}z) \frac{1}{\sqrt{\pi}} e^{-z^2} dz - \int_{x/2a\sqrt{t}}^{\infty} \varphi(-x + 2a\sqrt{t}z) \frac{1}{\sqrt{\pi}} e^{-z^2} dz \end{aligned}$$

其中, 可将 $f(t)$ 写成 $\int_0^t f'(\tau) d\tau + f(0)$, 则

$$\begin{aligned} & f(t) - \int_0^t f'(\tau) \operatorname{erf} \left(\frac{x}{2a\sqrt{t-\tau}} \right) d\tau - f(0) \operatorname{erf}(x / 2a\sqrt{t}) \\ &= \int_0^t f'(\tau) \left[1 - \operatorname{erf} \left(\frac{x}{2a\sqrt{t-\tau}} \right) \right] d\tau - f(0) [1 - \operatorname{erf}(x / 2a\sqrt{t})] \\ &= \int_0^t f'(\tau) \operatorname{erfc} \left(\frac{x}{2a\sqrt{t-\tau}} \right) d\tau + f(0) \operatorname{erfc}(x / 2a\sqrt{t}) \\ u(x, t) = & f(0) \operatorname{erfc} \left(\frac{x}{2a\sqrt{t}} \right) + \int_0^t f'(\tau) \operatorname{erfc} \left(\frac{x}{2a\sqrt{t-\tau}} \right) d\tau \\ & + \int_{-x/2a\sqrt{t}}^{\infty} \varphi(x + 2a\sqrt{t}z) \frac{1}{\sqrt{\pi}} e^{-z^2} dz - \int_{x/2a\sqrt{t}}^{\infty} \varphi(2a\sqrt{t}z - x) \frac{1}{\sqrt{\pi}} e^{-z^2} dz. \end{aligned}$$

11. 在一维半无界空间中求解 $u_t - a^2 u_{xx} = 0, u_x|_{x=0} = q(t), u|_{t=0} = 0$ 。

解：定解问题

$$\begin{cases} u_t - a^2 u_{xx} = 0, & (x > 0) \\ u_x|_{x=0} = q(t), \\ u|_{t=0} = 0. \end{cases} \quad (1)$$

先将边界条件齐次化，令 $u(x, t) = v(x, t) + w(x, t) - q(t)e^{-x}$ ，定解问题①转化成 v 和 w 的定解问题

$$\begin{cases} v_t - a^2 v_{xx} = q'(t)e^{-x} - a^2 q(t)e^{-x}, \\ w_x|_{x=0} = 0, \\ u|_{t=0} = 0. \end{cases} \quad (2)$$

$$\begin{cases} w_t - a^2 w_{xx} = 0, \\ w_x|_{x=0} = 0, \\ u|_{t=0} = q(0)e^{-x}. \end{cases} \quad (3)$$

根据边界条件，将定解问题②③偶延拓到的半空间

$$\begin{cases} v_t - a^2 v_{xx} = f(x, t) = \begin{cases} [q'(t) - a^2 q(t)]e^{-x}, & (x > 0) \\ [q'(t) - a^2 q(t)]e^x, & (x < 0) \end{cases} \\ v|_{t=0} = 0. \end{cases} \quad (4)$$

$$\begin{cases} w_t - a^2 w_{xx} = f(t) = 0, \\ w|_{t=0} = \varphi(x) = \begin{cases} q(0)e^{-x}, & (x > 0) \\ q(0)e^x, & (x < 0) \end{cases} \end{cases} \quad (5)$$

对④⑤作傅里叶变换，问题转化成

$$\begin{cases} V' + k^2 a^2 V = F(t, k), \\ V|_{t=0} = 0. \end{cases} \quad (6)$$

$$\begin{cases} W' + k^2 a^2 W = 0, \\ W|_{t=0} = \Phi(k). \end{cases} \quad (7)$$

(1) 先求解⑥这个非齐次常微分方程，用 $e^{k^2 a^2 t}$ 遍乘方程各项，得

$$\frac{d}{dt}[V(t; k)e^{k^2 a^2 t}] = F(t, k)e^{k^2 a^2 t}.$$

对 t 积分一次，并计及始值，

$$V(t; k)e^{k^2 a^2 t} = \int_0^t F(\tau, k)e^{k^2 a^2 \tau} d\tau,$$

即

$$\begin{aligned} V(t; k) &= e^{-k^2 a^2 t} \int_0^t F(\tau, k) e^{k^2 a^2 \tau} d\tau \\ &= \frac{1}{2\pi} \int_0^t e^{-k^2 a^2 (t-\tau)} \left[\int_{-\infty}^{\infty} f(\xi, \tau) e^{-ik\xi} d\xi \right] d\tau. \end{aligned}$$

进行逆傅里叶变换,

$$v(x, t) = \frac{1}{2\pi} \int_{-\infty}^{\infty} \left[\int_0^t e^{-k^2 a^2 (t-\tau)} \int_{-\infty}^{\infty} f(\xi, \tau) e^{-ik\xi} d\xi d\tau \right] e^{ikx} dk.$$

交换积分次序

$$v(x, t) = \frac{1}{2\pi} \int_0^t \int_{-\infty}^{\infty} f(\xi, \tau) \left[\int_{-\infty}^{\infty} e^{-k^2 a^2 (t-\tau)} e^{ik(x-\xi)} dk \right] d\xi d\tau.$$

引用教材本节例 2 的积分公式 $\int_{-\infty}^{\infty} e^{-\alpha^2 k^2} e^{\beta k} dk = (\sqrt{\pi}/\alpha) e^{\beta^2/4\alpha^2}$ 计算[]内的积分,

置 $\alpha = a\sqrt{t}, \beta = i(x-\xi)$, 以利用此积分公式, 即得

$$v(x, t) = \int_0^t \int_{-\infty}^{\infty} f(\xi, \tau) \left[\frac{1}{2a\sqrt{\pi(t-\tau)}} e^{-\frac{(x-\xi)^2}{4a^2(t-\tau)}} \right] d\xi d\tau.$$

代入 $f(x)$ 的具体表达式, 得到

$$\begin{aligned} v(x, t) &= \int_0^t \int_{-\infty}^0 [q'(t) - a^2 q(t)] e^{\xi} \left[\frac{1}{2a\sqrt{\pi(t-\tau)}} e^{-\frac{(x-\xi)^2}{4a^2(t-\tau)}} \right] d\xi d\tau \\ &\quad + \int_0^t \int_0^{\infty} [q'(t) - a^2 q(t)] e^{-\xi} \left[\frac{1}{2a\sqrt{\pi(t-\tau)}} e^{-\frac{(x-\xi)^2}{4a^2(t-\tau)}} \right] d\xi d\tau \end{aligned} \quad (8)$$

在 ⑧ 式右边第一个积分中令 $z = \frac{[x - 2a^2(t-\tau)] - \xi}{2a\sqrt{t-\tau}}$, 第二个积分中令

$z = \frac{\xi - [(x + 2a^2(t-\tau))]}{2a\sqrt{t-\tau}}$, 有

$$\begin{aligned} v(x, t) &= \frac{1}{2} \int_0^t [a^2 q(\tau) - q'(\tau)] \left\{ e^{\frac{4a^4(t-\tau)^2 + 4a^2 x(t-\tau)}{4a^2(t-\tau)}} \operatorname{erfc} \left[\frac{2a^2(t-\tau) + x}{2a\sqrt{t-\tau}} \right] \right. \\ &\quad \left. - e^{\frac{4a^4(t-\tau)^2 - 4a^2 x(t-\tau)}{4a^2(t-\tau)}} \operatorname{erfc} \left[\frac{2a^2(t-\tau) - x}{2a\sqrt{t-\tau}} \right] \right\} d\tau \\ &= \frac{1}{2} \int_0^t [a^2 q(\tau) - q'(\tau)] e^{a^2(t-\tau)} \left\{ e^x \operatorname{erfc} \left[\frac{2a^2(t-\tau) + x}{2a\sqrt{t-\tau}} \right] \right. \\ &\quad \left. - e^{-x} \operatorname{erfc} \left[\frac{2a^2(t-\tau) - x}{2a\sqrt{t-\tau}} \right] \right\} d\tau. \end{aligned}$$

(2) 再求解⑦

$$\begin{cases} W' + k^2 a^2 W = 0, \\ W|_{t=0} = \Phi(k). \end{cases}$$

$$W = \Phi(k) e^{-k^2 a^2 t} = \frac{e^{-k^2 a^2 t}}{2\pi} \int_{-\infty}^{\infty} \varphi(\xi) e^{-ik\xi} d\xi$$

进行逆傅里叶变换,

$$w(x, t) = \frac{1}{2\pi} \int_{-\infty}^{\infty} \left[\int_{-\infty}^{\infty} \varphi(\xi) e^{-ik\xi} d\xi e^{-k^2 a^2 t} \right] e^{ikx} dk.$$

交换积分次序

$$w(x, t) = \frac{1}{2\pi} \int_{-\infty}^{\infty} \varphi(\xi) \left[\int_{-\infty}^{\infty} e^{-k^2 a^2 t} e^{ik(x-\xi)} dk \right] d\xi.$$

引用教材本节例 2 的积分公式 $\int_{-\infty}^{\infty} e^{-\alpha^2 k^2} e^{\beta k} dk = (\sqrt{\pi} / \alpha) e^{\beta^2 / 4\alpha^2}$ 计算[]内的积分,

置 $\alpha = a\sqrt{t}, \beta = i(x - \xi)$, 以利用此积分公式, 即得

$$w(x, t) = \int_{-\infty}^{\infty} \varphi(\xi) \left[\frac{1}{2a\sqrt{\pi t}} e^{-\frac{(x-\xi)^2}{4a^2 t}} \right] d\xi.$$

代入 $\varphi(t)$ 的具体表达式, 得到

$$\begin{aligned} w(x, t) &= \int_{-\infty}^0 q(0) e^{\xi} \left[\frac{1}{2a\sqrt{\pi t}} e^{-\frac{(x-\xi)^2}{4a^2 t}} \right] d\xi + \int_0^{\infty} q(0) e^{-\xi} \left[\frac{1}{2a\sqrt{\pi t}} e^{-\frac{(x-\xi)^2}{4a^2 t}} \right] d\xi \\ &= \frac{1}{2} q(0) \left[e^{a^2 t + x} \operatorname{erfc} \left(\frac{2a^2 t + x}{2a\sqrt{t}} \right) - e^{a^2 t - x} \operatorname{erfc} \left(\frac{2a^2 t - x}{2a\sqrt{t}} \right) \right] \\ &= \frac{1}{2} q(0) e^{a^2 t} \left[e^x \operatorname{erfc} \left(\frac{2a^2 t + x}{2a\sqrt{t}} \right) - e^{-x} \operatorname{erfc} \left(\frac{2a^2 t - x}{2a\sqrt{t}} \right) \right]. \end{aligned}$$

从而, 问题的解是

$$\begin{aligned} u(x, t) &= v(x, t) + w(x, t) - q(t) e^{-x} \\ &= -q(t) e^{-x} + \frac{1}{2} q(0) e^{a^2 t} \left[e^x \operatorname{erfc} \left(\frac{2a^2 t + x}{2a\sqrt{t}} \right) - e^{-x} \operatorname{erfc} \left(\frac{2a^2 t - x}{2a\sqrt{t}} \right) \right] \\ &\quad + \frac{1}{2} \int_0^t [a^2 q(\tau) - q'(\tau)] e^{a^2(t-\tau)} \left\{ e^x \operatorname{erfc} \left[\frac{2a^2(t-\tau) + x}{2a\sqrt{t-\tau}} \right] \right. \\ &\quad \left. - e^{-x} \operatorname{erfc} \left[\frac{2a^2(t-\tau) - x}{2a\sqrt{t-\tau}} \right] \right\} d\tau. \end{aligned}$$

12. 例 7 研究三维无界空间中的受迫振动是从初始 ($t = 0$) 状态推算以后 ($t > 0$) 的状态。试重新求解例 7, 从初始 ($t = 0$) 状态反推以前 ($t < 0$) 的状态。

解: 定解问题为

$$\begin{cases} u_{tt} - a^2 \Delta_3 u = f(\mathbf{r}, t), \\ u|_{t=0} = 0, \quad u_t|_{t=0} = 0. \end{cases} \quad (1)$$

对①作傅里叶变换, 问题①变换为非齐次常微分方程的初始值问题

$$\begin{cases} U'' + k^2 a^2 U = F(t; \mathbf{k}), \\ U|_{t=0} = 0, \quad U'|_{t=0} = 0. \end{cases}$$

这个问题的解是 (参见 § 6.3 习题 7 答案)

$$U(t; \mathbf{k}) = \frac{1}{2a i k} \int_0^t F(\tau; \mathbf{k}) [e^{i k a (t-\tau)} - e^{-i k a (t-\tau)}] d\tau.$$

然后对 $U(t; \mathbf{k})$ 进行逆傅里叶变换.

$$\begin{aligned} u(\mathbf{r}, t) &= \iiint_{-\infty}^{\infty} \frac{1}{2a i k} \int_0^t F(\tau; \mathbf{k}) [e^{i k a (t-\tau)} - e^{-i k a (t-\tau)}] e^{i \mathbf{k} \cdot \mathbf{r}} d\tau d\mathbf{k}_1 d\mathbf{k}_2 d\mathbf{k}_3 \\ &= \frac{1}{4\pi a} \iiint_{-\infty}^{\infty} \int_0^t f(\mathbf{r}', \tau) \frac{1}{(2\pi)^3} \iiint_{-\infty}^{\infty} \frac{2\pi}{i k} [e^{i k a (t-\tau)} - e^{-i k a (t-\tau)}] e^{i \mathbf{k} \cdot (\mathbf{r} - \mathbf{r}')} d\mathbf{k}_1 d\mathbf{k}_2 d\mathbf{k}_3 dV'. \end{aligned}$$

引用 § 5.3 例 1 的结果, 并应用延迟定理,

$$u(\mathbf{r}, t) = \frac{1}{4\pi a} \iiint_{-\infty}^{\infty} \int_0^t f(\mathbf{r}', \tau) \frac{1}{|\mathbf{r} - \mathbf{r}'|} \{ \delta[|\mathbf{r} - \mathbf{r}'| - a(t - \tau)] + \delta[|\mathbf{r} - \mathbf{r}'| + a(t - \tau)] \} d\tau dV'.$$

再引用 (5.3.5) 以及关系式 $\delta(ax) = \delta(x)/|a|$,

$$\begin{aligned} u(\mathbf{r}, t) &= \frac{1}{4\pi a} \iiint_{-\infty}^{\infty} \int_0^t f(\mathbf{r}', \tau) \frac{1}{|\mathbf{r} - \mathbf{r}'|} \left[\delta\left(t - \tau - \frac{|\mathbf{r} - \mathbf{r}'|}{a}\right) + \delta\left(t - \tau + \frac{|\mathbf{r} - \mathbf{r}'|}{a}\right) \right] d\tau dV' \\ &= \frac{1}{4\pi a^2} \iiint_{-\infty}^{\infty} \frac{f(\mathbf{r}', t - |\mathbf{r} - \mathbf{r}'|/a)}{|\mathbf{r} - \mathbf{r}'|} dV' + \frac{1}{4\pi a^2} \iiint_{-\infty}^{\infty} \frac{f(\mathbf{r}', t + |\mathbf{r} - \mathbf{r}'|/a)}{|\mathbf{r} - \mathbf{r}'|} dV' \end{aligned}$$

其中, 右边第一项对应 $t > 0$, 表示是从初始 ($t = 0$) 状态推算以后 ($t > 0$) 的状态, 称为称为**推迟势**。第二项对应 $t < 0$, 表示从初始状态反推以前 ($t < 0$) 的状态, 称为**超前势**。合起来, 可写为

$$u(\mathbf{r}, t) = \frac{1}{4\pi a^2} \iiint_{-\infty}^{\infty} \frac{f(\mathbf{r}', |t| - |\mathbf{r} - \mathbf{r}'|/a)}{|\mathbf{r} - \mathbf{r}'|} dV'.$$

上面这个积分其实不必在无界空间进行, 只需在条件 $|t| - |\mathbf{r} - \mathbf{r}'|/a \geq 0$ 下积分. 换句话说, 对 $\mathbf{r}'$ 的积分只需在球体 $T_{a|t|}^{\mathbf{r}}$ 中进行, 此球的球心为径矢 $\mathbf{r}$, 而半径为 $a|t|$ 。这样,

$$u(\mathbf{r}, t) = \frac{1}{4\pi a^2} \iiint_{T_{a|t|}^{\mathbf{r}}} \frac{f(\mathbf{r}', |t| - |\mathbf{r} - \mathbf{r}'|/a)}{|\mathbf{r} - \mathbf{r}'|} dV'.$$

若仅考虑 $t < 0$ 的情形, 解可写为

$$\begin{aligned} u(\mathbf{r}, t) &= \frac{1}{4\pi a^2} \iiint_{T_{a|t|}^{\mathbf{r}}} \frac{f(\mathbf{r}', t + |\mathbf{r} - \mathbf{r}'|/a)}{|\mathbf{r} - \mathbf{r}'|} dV' \\ &= \frac{1}{4\pi a^2} \iiint_{T_{a|t|}^{\mathbf{r}}} \frac{f(\mathbf{r}', t + R/a)}{R} dV'. \end{aligned}$$

13. 试用降维法由泊松公式 (13.1.4) 推出一维波动的达朗贝尔公式 (7.4.7)。

$$\text{解: } u(\mathbf{r}, t) = \frac{1}{4\pi a} \frac{\partial}{\partial t} \iint_{S_{at}^r} \frac{\varphi(\mathbf{r}')}{at} dS' + \frac{1}{4\pi a} \iint_{S_{at}^r} \frac{\psi(\mathbf{r}')}{at} dS', \quad (13.1.4)$$

所谓一维波动，即沿一个方向的平面波动，其实还是在三维空间中传播的波动，这样说来，一维无界空间中的波动问题的解也由泊松公式 (13.1.4) 给出。但既然问题跟坐标 y 和 z 无关，当然不希望泊松公式中出现 y 和 z 。三维波动的泊松公式，消除了坐标 y 和 z ，就成为一维波动的公式，即达朗贝尔公式。对于一维波动，球面 S_{at}^r 上的积分应代之以沿 x 轴的一系列环面 $\Sigma_{at}^{x,y}$ 上的积分。参看图 13-4, 环面 $\Sigma_{at}^{x,y}$ 面积元

$$dS' = 2\pi(at \cdot \cos \theta) \cdot (at \cdot d\theta)$$

$$dx' = (at \cdot d\theta) \cos \theta$$

$$(at \cdot d\theta) = \frac{dx'}{\cos \theta}$$

$$dS' = 2\pi(at) dx'$$

即

$$\begin{aligned} u(\mathbf{r}, t) &= \frac{1}{4\pi a} \frac{\partial}{\partial t} \int_{x-at}^{x+at} \frac{\varphi(x')}{at} 2\pi(at) dx' + \frac{1}{4\pi a} \int_{x-at}^{x+at} \frac{\psi(x')}{at} 2\pi(at) dx' \\ &= \frac{1}{2a} \frac{\partial}{\partial t} \int_{x-at}^{x+at} \varphi(x') dx' + \frac{1}{2a} \int_{x-at}^{x+at} \psi(x') dx' \\ &= \frac{1}{2} [\varphi(x+at) + \varphi(x-at)] + \frac{1}{2a} \int_{x-at}^{x+at} \psi(x') dx'. \end{aligned} \quad (7.4.7)$$

§ 13.2 拉普拉斯变换法

1. 求解一维无界空间中的扩散问题 $u_t - a^2 u_{xx} = 0, u|_{t=0} = \varphi(x)$ 。

解：定解问题

$$\begin{cases} u_t - a^2 u_{xx} = 0, & (-\infty < x < \infty) \\ u|_{t=0} = \varphi(x). \end{cases} \quad (1)$$

对定解问题①施行拉普拉斯变换，得

$$p\bar{u} - \varphi(x) - a^2 \bar{u}_{xx} = 0,$$

即

$$\bar{u}_{xx} - \frac{p}{a^2} \bar{u} = -\frac{1}{a^2} \varphi(x), \quad (2)$$

以参数变异法求解②，与②对应的齐次方程的解是

$$\bar{u}_0 = A e^{\frac{\sqrt{p}}{a} x} + B e^{-\frac{\sqrt{p}}{a} x}, \quad (3)$$

③中的系数 A 和 B 是积分常数。为求得非齐次方程②的一个特解，将③中的系数 A 和 B 看作 x 的函数 $A(x)$ 和 $B(x)$ ，即设特解为

$$\bar{u}_1 = A(x) e^{\frac{\sqrt{p}}{a} x} + B(x) e^{-\frac{\sqrt{p}}{a} x}. \quad (4)$$

将④代入②

$$\bar{u}_1' = \frac{\sqrt{p}}{a} A(x) e^{\frac{\sqrt{p}}{a} x} - \frac{\sqrt{p}}{a} B(x) e^{-\frac{\sqrt{p}}{a} x} + A'(x) e^{\frac{\sqrt{p}}{a} x} + B'(x) e^{-\frac{\sqrt{p}}{a} x}, \quad (5)$$

引入辅助条件

$$A'(x) e^{\frac{\sqrt{p}}{a} x} + B'(x) e^{-\frac{\sqrt{p}}{a} x} = 0. \quad (6)$$

⑤式成为

$$\bar{u}_1' = \frac{\sqrt{p}}{a} A(x) e^{\frac{\sqrt{p}}{a} x} - \frac{\sqrt{p}}{a} B(x) e^{-\frac{\sqrt{p}}{a} x}, \quad (7)$$

$$\bar{u}_1'' = \frac{p}{a^2} \left[A(x) e^{\frac{\sqrt{p}}{a} x} + B(x) e^{-\frac{\sqrt{p}}{a} x} \right] + \frac{\sqrt{p}}{a} \left[A'(x) e^{\frac{\sqrt{p}}{a} x} - \frac{\sqrt{p}}{a} B'(x) e^{-\frac{\sqrt{p}}{a} x} \right], \quad (8)$$

以⑧代入②，得

$$\frac{\sqrt{p}}{a} \left[A'(x) e^{\frac{\sqrt{p}}{a} x} - \frac{\sqrt{p}}{a} B'(x) e^{-\frac{\sqrt{p}}{a} x} \right] = -\frac{1}{a^2} \varphi(x)$$

即

$$A'(x) e^{\frac{\sqrt{p}}{a} x} - \frac{\sqrt{p}}{a} B'(x) e^{-\frac{\sqrt{p}}{a} x} = -\frac{1}{a\sqrt{p}} \varphi(x). \quad (9)$$

⑥⑨联立求解，得到

$$\begin{cases} A'(x) = -\frac{1}{2a\sqrt{p}}\varphi(x)e^{-\frac{\sqrt{p}}{a}x} \\ B'(x) = \frac{1}{2a\sqrt{p}}\varphi(x)e^{\frac{\sqrt{p}}{a}x} \end{cases}$$

积分得

$$\begin{cases} A = -\frac{1}{2a\sqrt{p}}\int^{(x)}\varphi(\xi)e^{-\frac{\sqrt{p}}{a}\xi}d\xi \\ B = \frac{1}{2a\sqrt{p}}\int^{(x)}\varphi(\xi)e^{\frac{\sqrt{p}}{a}\xi}d\xi \end{cases}$$

以此代入④，即得方程②的特解为

$$\bar{u}_1 = -\frac{1}{2a\sqrt{p}}e^{\frac{\sqrt{p}}{a}x}\int^{(x)}\varphi(\xi)e^{-\frac{\sqrt{p}}{a}\xi}d\xi + \frac{1}{2a\sqrt{p}}e^{-\frac{\sqrt{p}}{a}x}\int^{(x)}\varphi(\xi)e^{\frac{\sqrt{p}}{a}\xi}d\xi,$$

从而得到方程②的通解为

$$\begin{aligned} \bar{u} &= Ae^{\frac{\sqrt{p}}{a}x} + Be^{-\frac{\sqrt{p}}{a}x} \\ &\quad - \frac{1}{2a\sqrt{p}}e^{\frac{\sqrt{p}}{a}x}\int^{(x)}\varphi(\xi)e^{-\frac{\sqrt{p}}{a}\xi}d\xi \\ &\quad + \frac{1}{2a\sqrt{p}}e^{-\frac{\sqrt{p}}{a}x}\int^{(x)}\varphi(\xi)e^{\frac{\sqrt{p}}{a}\xi}d\xi, \end{aligned}$$

系数 A 、 B 及积分下限由自然边界条件确定。要求 $x \rightarrow +\infty$ 解有限，置 $A=0$ ，同时第一个积分下限取为 $+\infty$ ；要求 $x \rightarrow -\infty$ 解有限，置 $B=0$ ，同时第二个积分下限取为 $-\infty$ ，从而

$$\begin{aligned} \bar{u} &= -\frac{1}{2a\sqrt{p}}e^{\frac{\sqrt{p}}{a}x}\int_{\infty}^x\varphi(\xi)e^{-\frac{\sqrt{p}}{a}\xi}d\xi \\ &\quad + \frac{1}{2a\sqrt{p}}e^{-\frac{\sqrt{p}}{a}x}\int_{-\infty}^x\varphi(\xi)e^{\frac{\sqrt{p}}{a}\xi}d\xi, \end{aligned}$$

即

$$\begin{aligned} \bar{u} &= \frac{1}{2a\sqrt{p}}\int_x^{\infty}\varphi(\xi)e^{-\frac{\sqrt{p}}{a}(\xi-x)}d\xi \\ &\quad + \frac{1}{2a\sqrt{p}}\int_{-\infty}^x\varphi(\xi)e^{-\frac{\sqrt{p}}{a}(x-\xi)}d\xi, \end{aligned}$$

进行反演，利用教材第 394 页公式 13， $\frac{e^{-a\sqrt{p}}}{\sqrt{p}} \doteq \frac{1}{\sqrt{\pi t}}e^{-a^2/4t}$ ，有

$$\frac{e^{\frac{(x-\xi)}{a}\sqrt{p}}}{\sqrt{p}} \doteq \frac{1}{\sqrt{\pi t}}e^{-(x-\xi)^2/4a^2t}$$

$$\frac{e^{-\frac{(\xi-x)}{a}\sqrt{p}}}{\sqrt{p}} \doteq \frac{1}{\sqrt{\pi t}}e^{-(\xi-x)^2/4a^2t} = \frac{1}{\sqrt{\pi t}}e^{-(x-\xi)^2/4a^2t}$$

故

$$\begin{aligned} u(x,t) &= \int_x^{\infty}\varphi(\xi)\left[\frac{1}{2a\sqrt{\pi t}}e^{-\frac{(x-\xi)^2}{4a^2t}}\right]d\xi + \int_{-\infty}^x\varphi(\xi)\left[\frac{1}{2a\sqrt{\pi t}}e^{-\frac{(x-\xi)^2}{4a^2t}}\right]d\xi \\ &= \int_{-\infty}^{\infty}\varphi(\xi)\left[\frac{1}{2a\sqrt{\pi t}}e^{-\frac{(x-\xi)^2}{4a^2t}}\right]d\xi. \end{aligned}$$

2. 求解硅片的限定源扩散问题.将硅片的厚度当作无限大,这是半无界空间的定解问题
 $u_t - a^2 u_{xx} = 0, u_x|_{x=0} = 0, u|_{t=0} = \Phi_0 \delta(x-0)$ 。[本题即 § 13.1 例 4, 可对照]

解: 定解问题

$$\begin{cases} u_t - a^2 u_{xx} = 0, \\ u_x|_{x=0} = 0, \\ u|_{t=0} = \Phi_0 \delta(x-0). \end{cases} \quad (1)$$

解法一: 对定解问题①施行拉普拉斯变换, 得

$$\begin{cases} p\bar{u} - \Phi_0 \delta(x-0) - a^2 \bar{u}_{xx} = 0, & (x>0) \\ \bar{u}_x|_{x=0} = 0. \end{cases} \quad (2)$$

由上题可知, ②的通解为

$$\begin{aligned} \bar{u} &= A e^{\frac{\sqrt{p}}{a}x} + B e^{-\frac{\sqrt{p}}{a}x} \\ &\quad - \frac{1}{2a\sqrt{p}} e^{\frac{\sqrt{p}}{a}x} \int_{(x)}^{\infty} \Phi_0 \delta(\xi-0) e^{-\frac{\sqrt{p}}{a}\xi} d\xi \\ &\quad + \frac{1}{2a\sqrt{p}} e^{-\frac{\sqrt{p}}{a}x} \int_{-\infty}^{(x)} \Phi_0 \delta(\xi-0) e^{\frac{\sqrt{p}}{a}\xi} d\xi, \end{aligned}$$

所以, $x \rightarrow +\infty$ 时, $\bar{u}$ 有界, 因此置 $A=0$ 。为保证积分收敛, 第一个积分下限取为 $+\infty$, 第二个积分下限取为 $-\infty$, 从而

$$\begin{aligned} \bar{u} &= B e^{-\frac{\sqrt{p}}{a}x} - \frac{1}{2a\sqrt{p}} e^{\frac{\sqrt{p}}{a}x} \int_{\infty}^x \Phi_0 \delta(\xi-0) e^{-\frac{\sqrt{p}}{a}\xi} d\xi \\ &\quad + \frac{1}{2a\sqrt{p}} e^{-\frac{\sqrt{p}}{a}x} \int_{-\infty}^x \Phi_0 \delta(\xi-0) e^{\frac{\sqrt{p}}{a}\xi} d\xi, \end{aligned}$$

又, 将解代入边界条件

$$\begin{aligned} \bar{u}_x &= -\frac{\sqrt{p}}{a} B e^{-\frac{\sqrt{p}}{a}x} - \frac{1}{2a} \frac{1}{\sqrt{p}} \delta(x-0) + \frac{1}{2a} \frac{1}{\sqrt{p}} \delta(x-0) = -\frac{\sqrt{p}}{a} B e^{-\frac{\sqrt{p}}{a}x}, \\ \bar{u}_x|_{x=0} &= -\frac{\sqrt{p}}{a} B e^{-\frac{\sqrt{p}}{a}x} - \frac{\sqrt{p}}{a} B e^{-\frac{\sqrt{p}}{a}x} \Big|_{x=0} = 0, \end{aligned}$$

有 $B=0$ 。于是

$$\begin{aligned} \bar{u} &= -\frac{1}{2a\sqrt{p}} e^{\frac{\sqrt{p}}{a}x} \int_{\infty}^x \Phi_0 \delta(\xi-0) e^{-\frac{\sqrt{p}}{a}\xi} d\xi \\ &\quad + \frac{1}{2a\sqrt{p}} e^{-\frac{\sqrt{p}}{a}x} \int_{-\infty}^x \Phi_0 \delta(\xi-0) e^{\frac{\sqrt{p}}{a}\xi} d\xi \\ &= \frac{\Phi_0}{2a\sqrt{p}} e^{\frac{\sqrt{p}}{a}x} + \frac{\Phi_0}{2a\sqrt{p}} e^{-\frac{\sqrt{p}}{a}x} \end{aligned}$$

反演, 得

$$u(x,t) = \frac{\Phi_0}{2a\sqrt{\pi t}} e^{-\frac{x^2}{4a^2t}} + \frac{\Phi_0}{2a\sqrt{\pi t}} e^{-\frac{x^2}{4a^2t}} = \frac{\Phi_0}{a\sqrt{\pi t}} e^{-\frac{x^2}{4a^2t}}。$$

解法二: 根据边界条件 $u_x|_{x=0} = 0$, 作偶延拓,

$$\begin{cases} u_t - a^2 u_{xx} = 0, \\ u_x|_{x=0} = 0, \\ u|_{t=0} = \begin{cases} \Phi_0 \delta(x-0), \\ \Phi_0 \delta(x+0). \end{cases} \end{cases}$$

直接引用 § 13.1 例 4 结果, 有

$$\begin{aligned} u(x,t) &= \frac{1}{2a} \int_{-\infty}^{\infty} \frac{\Phi_0 \delta(x-0) + \Phi_0 \delta(x+0)}{\sqrt{\pi t}} e^{-\frac{(\xi-x)^2}{4a^2t}} d\xi \\ &= \frac{\Phi_0}{2a\sqrt{\pi t}} \left[e^{-\frac{(0-x)^2}{4a^2t}} + e^{-\frac{(-0-x)^2}{4a^2t}} \right] \\ &= \frac{\Phi_0}{a\sqrt{\pi t}} e^{-\frac{x^2}{4a^2t}}。 \end{aligned}$$

3. 求解一维无界空间的有源输运问题 $u_t - a^2 u_{xx} = f(x, t)$, $u|_{t=0} = 0$ 。[本题即 § 13.1 例 3, 可对照]

解: 定解问题

$$\begin{cases} u_t - a^2 u_{xx} = f(x, t), & (-\infty < x < \infty) \\ u|_{t=0} = 0. \end{cases} \quad (1)$$

对定解问题①施行拉普拉斯变换, 得

$$p\bar{u} - a^2 \bar{u}_{xx} = \bar{f}(x, p),$$

应用第 1 题的结果

$$\begin{aligned} \bar{u} &= A e^{\frac{\sqrt{p}}{a}x} + B e^{-\frac{\sqrt{p}}{a}x} \\ &\quad - \frac{1}{2a} e^{\frac{\sqrt{p}}{a}x} \int^{(x)} \bar{f}(\xi, p) \frac{1}{\sqrt{p}} e^{-\frac{\sqrt{p}}{a}\xi} d\xi \\ &\quad + \frac{1}{2a} e^{-\frac{\sqrt{p}}{a}x} \int^{(x)} \bar{f}(\xi, p) \frac{1}{\sqrt{p}} e^{\frac{\sqrt{p}}{a}\xi} d\xi, \end{aligned}$$

自然边界条件要求 $x \rightarrow \pm\infty$ 解有限, 因此置 $A=B=0$, 同时, 第一个积分下限取为 $+\infty$, 第二个积分下限取为 $-\infty$, 从而

$$\bar{u} = -\frac{1}{2a} e^{\frac{\sqrt{p}}{a}x} \int_{\infty}^x \bar{f}(\xi, p) \frac{1}{\sqrt{p}} e^{-\frac{\sqrt{p}}{a}\xi} d\xi + \frac{1}{2a} e^{-\frac{\sqrt{p}}{a}x} \int_{-\infty}^x \bar{f}(\xi, p) \frac{1}{\sqrt{p}} e^{\frac{\sqrt{p}}{a}\xi} d\xi,$$

即

$$\bar{u} = \frac{1}{2a} \int_x^{\infty} \bar{f}(\xi, p) \frac{1}{\sqrt{p}} e^{-\frac{\sqrt{p}}{a}(\xi-x)} d\xi + \frac{1}{2a} \int_{-\infty}^x \bar{f}(\xi, p) \frac{1}{\sqrt{p}} e^{-\frac{\sqrt{p}}{a}(x-\xi)} d\xi,$$

进行反演

$$\bar{f}(\xi, p) = f(x, t), \quad \frac{e^{\frac{(x-\xi)\sqrt{p}}{a}}}{\sqrt{p}} = \frac{1}{\sqrt{\pi t}} e^{-(x-\xi)^2/4a^2t}$$

再应用卷积定理, 得到结果

$$u(x, t) = \int_{\xi=-\infty}^{\infty} \int_{\tau=0}^t \frac{1}{2a\sqrt{\pi(t-\tau)}} e^{-(x-\xi)^2/4a^2(t-\tau)} f(\xi, \tau) d\xi d\tau.$$

4. 求解一维半无界空间的输运问题 $u_t - a^2 u_{xx} = 0, u|_{t=0} = 0$, 边界条件是 $u|_{x=0} = f(t)$ 。[本题为 § 13.1 习题 10 的一部分, 可对照。]

解: 定解问题

$$\begin{cases} u_t - a^2 u_{xx} = 0, & (x > 0) \\ u|_{x=0} = f(t), \\ u|_{t=0} = 0. \end{cases} \quad (1)$$

对定解问题①施行拉普拉斯变换, 得

$$\begin{cases} p\bar{u} - a^2 \bar{u}_{xx} = 0, \\ \bar{u}|_{x=0} = \bar{f}(p). \end{cases} \quad (2)$$

②的通解为

$$\bar{u} = A e^{\frac{\sqrt{p}}{a} x} + B e^{-\frac{\sqrt{p}}{a} x},$$

由条件 $x \rightarrow +\infty$ 解有限, 置 $A=0$, 这样

$$\bar{u} = B e^{-\frac{\sqrt{p}}{a} x},$$

又由②中边值可得, $B = \bar{f}(p)$,

$$\bar{u} = \bar{f}(p) e^{-\frac{\sqrt{p}}{a} x} = p \left[f(p) \frac{1}{p} e^{-\frac{\sqrt{p}}{a} x} \right],$$

进行反演, $\bar{f}(p) = f(t)$, $\frac{1}{p} e^{-\frac{\sqrt{p}}{a} x} = \operatorname{erfc}\left(\frac{x}{a\sqrt{t}}\right)$, 应用卷积定理

$$f(p) \frac{1}{p} e^{-\frac{\sqrt{p}}{a} x} = \int_0^t f(t-\tau) \operatorname{erfc}\left(\frac{x}{a\sqrt{\tau}}\right) d\tau,$$

上式积分在 $t=0$ 时为零, 于是按拉普拉斯变换导数定理 (6.1.12) 有

$$u(x, t) = \frac{\partial}{\partial t} \int_0^t f(t-\tau) \operatorname{erfc}\left(\frac{x}{2a\sqrt{\tau}}\right) d\tau.$$

5. 求解无界弦的受迫振动 $u_{tt} - a^2 u_{xx} = f(x, t), u|_{t=0} = \varphi(x), u_t|_{t=0} = \psi(x)$ 。

解：

$$\begin{cases} u_{tt} - a^2 u_{xx} = f(x, t), \\ u|_{t=0} = \varphi(x), u_t|_{t=0} = \psi(x). \end{cases} \quad (1)$$

对定解问题①施行拉普拉斯变换，得

$$p^2 \bar{u} - p\varphi(x) - \psi - a^2 \bar{u}_{xx} = f(x, p),$$

即

$$\bar{u}_{xx} - \frac{p^2}{a^2} \bar{u} = -\frac{1}{a^2} [p\varphi(x) + \psi(x) - \bar{f}(x, p)], \quad (2)$$

方程②的通解为

$$\begin{aligned} \bar{u} = & A e^{\frac{p}{a}x} + B e^{-\frac{p}{a}x} \\ & - \frac{1}{2a} e^{\frac{p}{a}x} \int_{(x)}^{\infty} \frac{1}{p} [p\varphi(\xi) + \psi(\xi) - \bar{f}(\xi, p)] e^{-\frac{p}{a}\xi} d\xi \\ & + \frac{1}{2a} e^{-\frac{p}{a}x} \int_{-\infty}^{(x)} \frac{1}{p} [p\varphi(\xi) + \psi(\xi) - \bar{f}(\xi, p)] e^{\frac{p}{a}\xi} d\xi, \end{aligned}$$

系数 A 、 B 及积分下限由自然边界条件确定。要求 $x \rightarrow +\infty$ 解有限，置 $A=0$ ，同时第一个积分下限取为 $+\infty$ ；要求 $x \rightarrow -\infty$ 解有限，置 $B=0$ ，同时第二个积分下限取为 $-\infty$ ，从而

$$\begin{aligned} \bar{u} = & -\frac{1}{2a} \int_{-\infty}^x \frac{1}{p} [p\varphi(\xi) + \psi(\xi) - \bar{f}(\xi, p)] e^{-\frac{p}{a}(\xi-x)} d\xi \\ & + \frac{1}{2a} \int_{-\infty}^x \frac{1}{p} [p\varphi(\xi) + \psi(\xi) - \bar{f}(\xi, p)] e^{-\frac{p}{a}(x-\xi)} d\xi \\ = & \left[\frac{1}{2a} \int_x^{\infty} \frac{1}{p} \cdot \psi(\xi) e^{-\frac{p}{a}(\xi-x)} d\xi + \frac{1}{2a} \int_{-\infty}^x \frac{1}{p} \cdot \psi(\xi) e^{-\frac{p}{a}(x-\xi)} d\xi \right] \\ & + \left[\frac{1}{2a} \int_x^{\infty} \frac{1}{p} \cdot p\varphi(\xi) e^{-\frac{p}{a}(\xi-x)} d\xi + \frac{1}{2a} \int_{-\infty}^x \frac{1}{p} \cdot p\varphi(\xi) e^{-\frac{p}{a}(x-\xi)} d\xi \right] \\ & + \left[\frac{1}{2a} \int_x^{\infty} \frac{1}{p} \cdot \bar{f}(\xi, p) e^{-\frac{p}{a}(\xi-x)} d\xi + \frac{1}{2a} \int_{-\infty}^x \frac{1}{p} \cdot \bar{f}(\xi, p) e^{-\frac{p}{a}(x-\xi)} d\xi \right], \end{aligned}$$

再进行反演

$$\begin{aligned} \frac{1}{p} e^{-p \frac{(\xi-x)}{a}} & \doteq H\left(t - \frac{\xi-x}{a}\right) = \begin{cases} 1, & (\xi < x+at) \\ 0, & (\xi > x+at) \end{cases} \\ \frac{1}{p} e^{-p \frac{(x-\xi)}{a}} & \doteq H\left(t - \frac{x-\xi}{a}\right) = \begin{cases} 1, & (\xi > x-at) \\ 0, & (\xi < x-at) \end{cases} \\ \int_x^{\infty} \frac{1}{p} \cdot \psi(\xi) e^{-\frac{p}{a}(\xi-x)} d\xi + \int_{-\infty}^x \frac{1}{p} \cdot \psi(\xi) e^{-\frac{p}{a}(x-\xi)} d\xi & \doteq \int_{x-at}^{x+at} \psi(\xi) d\xi \end{aligned}$$

$$\begin{aligned} & p \left[\int_x^{\infty} \frac{1}{p} \varphi(\xi) e^{-\frac{p}{a}(\xi-x)} d\xi + \int_{-\infty}^x \frac{1}{p} \varphi(\xi) e^{-\frac{p}{a}(x-\xi)} d\xi \right] \\ & \doteq \frac{\partial}{\partial t} \int_{x-at}^{x+at} \varphi(\xi) d\xi = a [\varphi(x+at) + \varphi(x-at)] \\ & \int_x^{\infty} \frac{1}{p} \cdot \bar{f}(\xi, p) e^{-\frac{p}{a}(\xi-x)} d\xi + \int_{-\infty}^x \frac{1}{p} \cdot \bar{f}(\xi, p) e^{-\frac{p}{a}(x-\xi)} d\xi \\ & \doteq \int_0^t \int_{x-a(t-\tau)}^{x+a(t-\tau)} f(\xi, \tau) d\xi d\tau \end{aligned}$$

总的解是

$$u(x, t) = \frac{1}{2} [\varphi(x+at) + \varphi(x-at)] + \frac{1}{2} \int_{x-at}^{x+at} \psi(\xi) d\xi + \frac{1}{2a} \int_{\tau=0}^t \int_{\xi=x-a(t-\tau)}^{x+a(t-\tau)} f(\xi, \tau) d\xi d\tau.$$